

Direct Extraction of Sodium Hydroxide by Calix[4]pyrrole-Based Ion-Pair Receptors

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ABSTRACT: Described in this work are calix[4]pyrrole-based ion-pair receptors, *cis/trans*-1 and *cis/trans*-2, designed for the extraction of sodium hydroxide. An X-ray diffraction analysis of a single crystal of the *cis*-1·NaOH isomer isolated from a mixture of *cis/trans*-1 revealed a unique dimeric supramolecular structure. An average dimer in toluene-*d*₈ solution was inferred on the basis of diffusion-ordered spectroscopy (DOSY). Support for the proposed stoichiometry came from density functional theory (DFT) calculations. The structural stability of the dimeric *cis*-1·NaOH complex in toluene solution was further confirmed by ab initio molecular dynamics (AIMD) simulation with explicit representation of solvent. Under conditions of liquid-liquid extraction (LLE), purified receptors *cis*- and *trans*-2, were both found to remove NaOH from a pH 11.01 aqueous source phase into toluene with extraction efficiencies (E%) of 50%-60% when used equimolar to NaOH. However, in all cases precipitation was observed. Complexities associated with precipitation could be avoided by immobilization of the receptors onto a chemically inert poly(styrene) resin by means of solvent impregnation. The use of solvent impregnated resins (SIRs) eliminated precipitation in solution while retaining the extraction efficiency toward NaOH. This allowed both the pH and salinity of the alkaline source phase to be lowered.

Introduction

As a target for separations, sodium hydroxide receives comparatively little attention, yet over 80 million tonnes^{1,2} of this commodity chemical are produced and consumed by industry each year globally.³ Large-scale uses include, for example, the production of paper⁴ and alumina,^{5,6} along with chemicals, textiles, and myriad large, though smaller-scale processes.⁷ In most if not all of these applications, economics and environmental protection concerns mandate internal recycle. Ultimately, residual alkalinity is neutralized to eliminate its hazard. This produces salt or solid residues that require additional chemical inputs to manage the resulting waste streams. A specific niche problem, treating alkaline nuclear waste to remove caustic,^{8,9} has also attracted recent attention because of the immense cost of high-level-waste vitrification (hundreds of thousands of dollars per cubic meter) and the premium associated with reducing waste volumes. Better separations of sodium hydroxide would yield benefits in maximizing recycle and minimizing neutralization with acid. Proposed approaches have focused on converting hydroxide to other forms of alkalinity (e.g., alkoxides or phenoxides);^{8–10} however, in practice, bulk components are

typically precipitated or otherwise separated away from the NaOH. An attractive alternative would involve removing directly alkaline ion pairs, such as NaOH, from a high pH aqueous source phase.

To our knowledge, the challenge of direct NaOH extraction using an ion-pair receptor-based approach has yet to be met.¹¹ However, the problem of removing NaOH from basic media has received prior attention. For instance, weak hydroxy acids, such as lipophilic fluorinated alcohols and phenols, have been used to remove NaOH from alkaline source phases by exchanging H⁺ for Na⁺ under conditions of liquid-liquid extraction (LLE).^{8,9} It was found that the extraction efficiency could be enhanced by introducing cation host molecules, such as crown ethers, to promote synergistic H⁺ for Na⁺ exchange.¹² A reaction-based system that exploits the Lewis acidity of boronic esters was also shown effective towards hydroxide extraction.¹³ However, strategies that removes the OH[−] anion directly from an aqueous source phase without adding a Brønsted or Lewis acid remain underdeveloped. Here we report the synthesis and study of two calix[4]pyrrole-based ion-pair receptors *cis/trans*-1 and *cis/trans*-2 (Fig. 1) that function as NaOH capture agents.

Recognition and extraction of hydroxide salts is challenging due to the high basicity of the OH[−] anion.¹⁴ Moreover, the problem of removing NaOH directly from aqueous source phases is exacerbated by the high water solubility of both Na⁺ and OH[−] (hydration energies of −365, and −430 kJ mol^{−1}, respectively).^{15–18} Ion pair receptors may provide a means of overcoming these difficulties. They can enforce an increased electrostatic interaction between co-bound ions with a concomitant decrease in solvation and

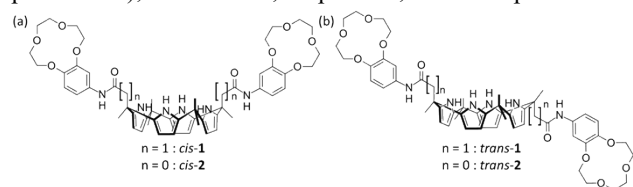
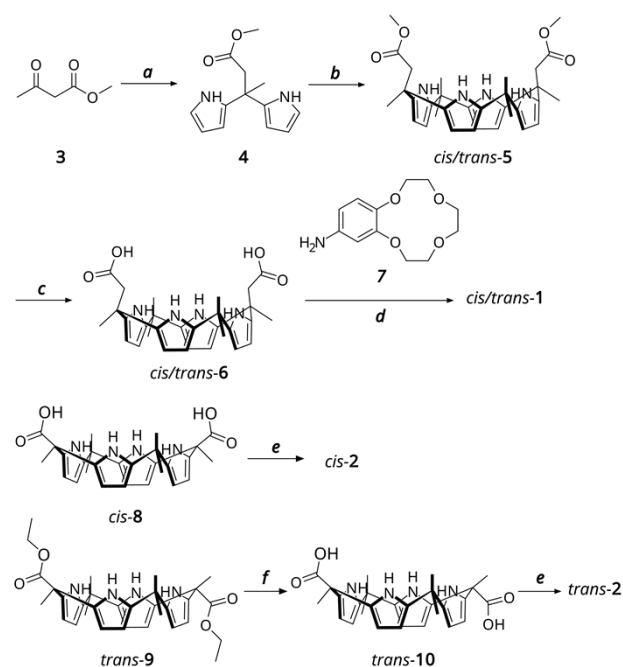


Figure 1. Chemical structures of (a) *cis/trans*-1 and (b) *cis/trans*-2.

Scheme 1. Synthesis of *cis/trans*-1 and *cis/trans*-2.^a



^aExperimental conditions: (a) pyrrole, TFA, acetone, rt, 0.5 h; (b) $\text{BF}_3 \cdot \text{OEt}_2$, acetone/MeOH, rt, 2 h; (c) KOH, THF/EtOH/ H_2O , 90 °C, 6 h; (d) 7, EDCI·HCl, pyridine, CH_2Cl_2 , rt, 6 h; (e) 7, EDCI·HCl, HOBT, DIPEA, anhydrous DMF, rt, 24 h; (f) KOH, THF/EtOH/ H_2O , 90 °C, 6 h.

often exhibit greater affinity and selectivity for target ion pairs than the corresponding single ion receptors.^{19–22} In 2017, our group described a hemispherand-strapped calix[4]pyrrole receptor capable of extracting CsOH and Cs_2CO_3 selectively over other alkaline hydroxide salts, including NaOH .²³ This led us to suggest that direct extraction of NaOH would necessitate re-engineering the core structure.

Previous research with crown ethers led to an appreciation that the stability of Na^+ in appropriately designed sandwich complexes is greater than in the corresponding monomeric complex due to the approximately two-fold increase in ion–dipole interactions.^{24,25} We thus envisioned that the conjugation of a calix[4]pyrrole core with two benzo-12-crown-4 subunits, which are known to form sandwich complexes with the Na^+ ion,^{26–32} would allow us to create an ion pair receptor capable of recognizing and extracting NaOH . The calix[4]pyrrole-based ion pair receptors *cis/trans*-1 and *cis/trans*-2 (Fig. 1) were designed to test this hypothesis.

Results and Discussion

Receptors *cis/trans*-1 and *cis/trans*-2 were synthesized according to Scheme 1. Methyl acetoacetate 3 was condensed with pyrrole in the presence of trifluoroacetic acid (TFA) to generate the ester-functionalized dipyrromethane 4 in 80% yield. Further condensation of the dipyrromethane 4 with acetone in the presence of $\text{BF}_3 \cdot \text{OEt}_2$ produced *cis/trans*-1 in 23% yield. Unfortunately, attempts to separate the *cis* and *trans* isomers 5 were not successful.

The ester groups of *cis/trans*-5 were saponified with KOH and then acidified with aqueous HCl to give the diacids *cis/trans*-6 in 88% yield. Coupling of *cis/trans*-6 with 7

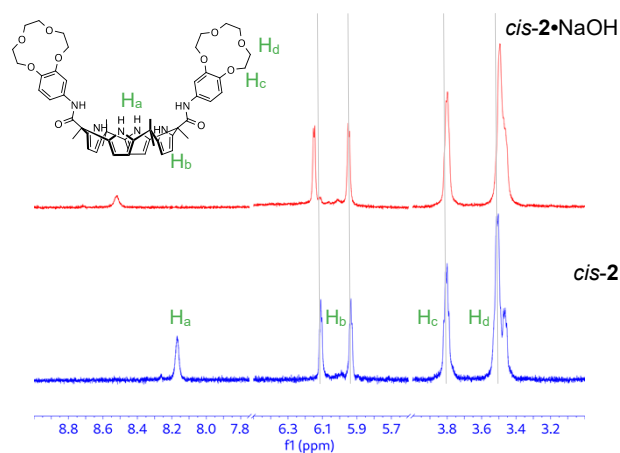


Figure 2. Partial ^1H NMR spectra of *cis*-2 (blue) and presumed *cis*-2·NaOH (red) in $\text{toluene-}d_8$. Samples were prepared as follows: Each receptor as a 1 mM solution in CD_3OD was mixed with 2 equiv of NaOH dissolved in CD_3OD . The CD_3OD was removed under high vacuum and the residue was taken up in $\text{toluene-}d_8$ and filtered through a cotton plug prior to recording the spectrum.

using EDCI·HCl and pyridine in methylene chloride afforded a mixture of *cis*-1 and *trans*-1 in 32% yield. Congeners *cis*-2 and *trans*-2, analogues lacking methylene bridges between the amide groups and the meso carbons of the calix[4]pyrrole core, were synthesized according to Scheme 1. Coupling of the readily isolated precursor *cis*-8³³ with 7 in the presence of EDCI·HCl, HOBT, and DIPEA in anhydrous DMF produced *cis*-2 in 47% yield. (Note: A list of abbreviations is provided at the end of this report.) Saponification of the reported compound *trans*-9³³ using KOH, followed by acidification with aqueous HCl, afforded the free acid, *trans*-10, in 93% yield. *Trans*-10 was coupled with 7 in the presence of EDCI·HCl, HOBT, and DIPEA in dry DMF. This gave the desired compound, *trans*-2 in 37% yield.

Since a key study goal was to obtain insights into the effects of geometrical isomerization, we elected to carry out our NaOH extraction studies with the purified isomers *cis*-2 and *trans*-2, rather than a mixture of *cis*-1/*trans*-1. Considering the only difference between receptors 1 and receptors 2 is a methylene linking unit, we think their characteristics, including ion binding and solubility, which are dominated by the crown and calix[4]pyrrole subunits, should be quite similar. Since the hydroxide anion reacts readily with chlorinated solvents,^{34,35} toluene was used for solution studies. NaOH binding by configurationally pure receptors *cis*-2 and *trans*-2 was first assessed by ^1H NMR spectroscopy in $\text{toluene-}d_8$. Upon contacting with NaOH dissolved in methanol, both receptors showed significant precipitation at 1 mM in the presence of equimolar NaOH, thus precluding accurate titration experiments. Nevertheless, filtered solutions of presumed *cis*-2·NaOH and *trans*-2·NaOH displayed appreciable changes in their respective ^1H NMR spectra relative to the free receptors (Fig 2, cf. Fig. S1). The singlet ascribed to the calix[4]pyrrole NH protons underwent a downfield shift when treated with 2 equivalents of NaOH ($\Delta\delta_{\text{Ha}} = 0.44$ and 0.62 ppm for *cis*-2 and *trans*-2, respectively). In both *cis*-2 and *trans*-2, the β -pyrrolic proton signals (H_b) exhibited slight downfield shifts. Note that amide protons were not clearly resolved for both receptors.

These changes in the chemical shifts are taken as evidence of a hydrogen bonding interaction between the bound OH⁻ and the calix[4]pyrrole NH protons. Compared to the free receptors *cis*-2 and *trans*-2, the crown ether methylene peaks of the presumed sodium complexes (H_a) exhibited slight upfield shifts. These chemical shifts mirror those seen in the sandwiched complexes of benzo-15-crown-5 subunits in response to K⁺ ion binding.^{36,37}

To our surprise, only single crystals of *cis*-1•NaOH were isolated upon subjecting a dichloromethane/methanol/acetone solution of *cis*/*trans*-1 (mixture of configurational isomers) to slow evaporation in the presence of excess NaOH. The resulting structure revealed two individual Na⁺ cations sandwiched between two benzo-12-crown-4-ether moieties and coordinated by all crown ether oxygen atoms with Na⁺⋯O (benzocrown ether) distances between 2.301 and 2.527 Å (Fig. 3). The OH⁻ anions are hydrogen-bonded to the NH protons of the calix[4]pyrrole subunits in their respective cone conformations as inferred from the N⋯OH⁻ distances of 2.743–2.786 Å. The distances between the Na⁺ cations and the OH⁻ anions were 11.122–11.456 Å, as would be expected for a host-separated ion pair complex.^{19,20} No evidence for the linker amide NH protons forming hydrogen bonds to the bound OH⁻ anion was found.

Pyrrole and typical amide bonds have pK_a values of ca. 23 and 21.5, respectively, in DMSO.^{38,39} Hydroxide and fluoride have about the same thermochemical radius. As naked anions, both are extremely strong bases and can deprotonate weak acids. However, when they are solvated in hydrogen-bond-donating media, their basicity is moderated. Calix[4]pyrrole “solvates” both anions well via 4 hydrogen bonds, and our group has observed very high binding constants with fluoride with no evidence of deprotonation of the C4P unit.^{33,40} As in the case of fluoride, hydroxide will be strongly bound with what we expect to be comparable binding constants. Given the expected very high pK_a for the pyrrole NH protons, binding is thermodynamically expected to win out over deprotonation. In fact, no evidence that the pyrrolic NH (or amide NH) protons are deprotonated under the crystallization conditions was seen from the structural analysis. To the best of our knowledge, this is the first example of NaOH being stabilized within a supramolecular structure by an organic receptor in the solid state.

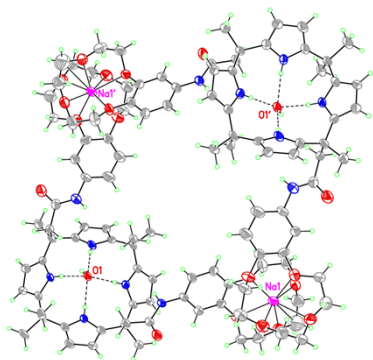


Figure 3. Single crystal X-ray diffraction structure of the sodium hydroxide complex [*cis*-1•NaOH]₂. Displacement ellipsoids are scaled to the 30% probability level. Dashed lines are indicative of presumed H-bonding interactions. The complex resides around a crystallographic inversion center at ½, ½, ½.

Efforts were made to determine whether the dimeric structure seen in the solid state dominates in solution. We thus performed diffusion-ordered spectroscopy (DOSY) to derive the size-dependent diffusion coefficient D. The diffusion coefficients were ca. $(9.68 \pm 0.26) \times 10^{-6}$ cm²/s and $(1.00 \pm 0.56) \times 10^{-5}$ cm²/s for *cis*-2 and *trans*-2, respectively, in toluene-*d*₈ (cf. Fig. S5-S6, Table 1). Smaller diffusion coefficient values of $(5.85 \pm 0.15) \times 10^{-6}$ and $(5.05 \pm 0.40) \times 10^{-6}$ were obtained after the addition of 2 equivalents of NaOH to solutions of *cis*-2 and *trans*-2, respectively (cf. Figs. S7-S8, Table 1). These results are consistent with the formation of larger aggregates in the presence of NaOH.

The molecular weights of the sodium hydroxide complexes of *cis*-2 or *trans*-2 were approximated according to equation 1:

$$D \propto \text{MW}^{-\alpha} \quad (1)$$

Table 1. Diffusion coefficients and estimated molecular weights of *cis*-2, *trans*-2, *cis*-2 + 2 equiv of NaOH and *trans*-2 + 2 equiv of NaOH.^a

0 equiv of NaOH		2 equiv of NaOH	
D (cm ² /s)	calcd. MW	D (cm ² /s)	calcd. MW
<i>cis</i> -2 $(9.68 \pm 0.26) \times 10^{-6}$	884 ± 66	$(5.85 \pm 0.15) \times 10^{-6}$	2039 ± 103
<i>trans</i> -2 $(1.00 \pm 0.56) \times 10^{-5}$	990 ± 138	$(5.05 \pm 0.40) \times 10^{-6}$	2842 ± 462

^aSamples were prepared as described in the caption to Figure 2. The D_{tol-*d*7} values of 2.27×10^{-5} , 2.47×10^{-5} , 1.90×10^{-5} , and 1.86×10^{-5} (cm²/s) were used as the internal reference for the samples of *cis*-2, *trans*-2, *cis*-2 + 2 equiv of NaOH, and *trans*-2 + 2 equiv of NaOH, respectively.

where D denotes the diffusion coefficient; MW denotes the molecular weight and α is a shape-dependent Flory exponent. While a ±30% error in estimating molecular weights by equation 1 has been reported,⁴¹ the results still provide insights into the solution structures. First, equation 1 was applied to estimate α given the known molecular weights of free *cis*-2, *trans*-2, and toluene-*d*₇ (partially deuterated; MW = 99) as an internal reference. We chose α = 0.39 in accord with the presumed globular geometry in solution.⁴² This choice also gave the lowest error relative to other α values tested. In these cases, molecular weights of 884 ± 66 g/mol and 990 ± 138 g/mol were derived, respectively (Table 1). These results are in agreement with the theoretical molecular weight for these isomers of 931 g/mol, thus providing a degree of confidence in the applicability of equation 1.

Considering the binding mode, a globular sandwich-like structure is expected for *cis*-2 in analogy to what is seen for *cis*-1•NaOH in the solid state. Thus, an α value of 0.39 was used to estimate the molecular weight of both *cis*-2•NaOH and *trans*-2•NaOH. However, we note that a number of assumptions are made here and that the resulting derived molecular weights should be considered with caution. Second, and subject to the above caveats, the approximate molecular weight for the filtrate of *cis*-2 + 2 equiv NaOH was 2039 ± 103 g/mol, which is close to the theoretical molecular weight of [*cis*-2•NaOH]₂ (MW = 1942 g/mol). In contrast, the approximate molecular weight of the filtrate from the *trans*-2 + 2 equiv of NaOH sample proved to be 2842 ± 462 g/mol, which is close to the molecular weight of [*trans*-2•NaOH]₃ (MW = 2913 g/mol). On this basis and considering the single crystal X-ray diffraction structure of [*cis*-1•NaOH]₂, we

propose *cis*-2 stabilizes an average 2:2 complex with NaOH. The assignment is less certain in the case of *trans*-2 where the DOSY data are most readily interpreted in terms of an average 3:3 complex with NaOH in solution. Artistic renditions of the possible structures are shown in Fig. S9.

Density functional theory (DFT) calculations at the B3LYP/6-31+G(d) level^{43,44} were performed to evaluate the relative stability of the 1:1 and 2:2 complexes of *cis*-1 and *cis*-2 with NaOH. Structures of the most stable complexes identified from multiple initial configurations are shown in Figures S39 and S40. They all show similar structural arrangements for the NaOH ion pair, in which the OH⁻ anion interacts with all four NH protons of the calix[4]pyrrole core and the Na⁺ cation forms eight short contacts with the oxygen donor atoms of the two crown ether moieties in a sandwiched configuration. The 2:2 complexes are enthalpically favored relative to two 1:1 complexes (*cf.* Table S2), both in the gas phase and in an IEFPCM⁴⁵ implicit solvent (by 16.5–17.1 kcal/mol for *cis*-1 and 13.4–25.5 kcal/mol for *cis*-2). The higher stability of the dimeric systems can be attributed to the flexibility of the host that is thus able to accommodate the NaOH guest in a 2:2 complex. Unfortunately, efforts to model a putative 3:3 complex proved unsuccessful. Accurate modeling of the 3:3 complex is not possible due to the size of the 3:3 complex and more importantly an inability to sample adequately the diversity of conformations possible for such a large ensemble. As such, it proved difficult to treat the 3:3 complex at the same level of DFT (B3LYP/6-31+G(d)) with high confidence in the results as is feasible for the 1:1 and 2:2 complexes.

In order to understand the microscopic behavior and structural stability of dimeric supramolecular structures in solution, ab initio molecular dynamics simulations (AIMD) employing the PBE-D3^{46,47} density function were carried out for [*cis*-1•NaOH]₂ in toluene. The simulation system was constructed by placing [*cis*-1•NaOH]₂ and 77 toluene molecules in a cubic box of length 26 Å. Periodic boundary conditions were employed in all three directions. The system was relaxed to an equilibrium state using classical force fields before performing the production runs using AIMD. The complex remained intact over the course of AIMD simulations (15 ps), with the OH⁻ and Na⁺ ions preserving their primary coordination to the pyrrole N-H groups and 12-crown-4 ether oxygen atoms, respectively. The average contact distances obtained from radial distribution functions (*cf.* Figure S41) match well with those in the crystal structure (the difference is 0.14 Å for Na···O and 0.017 Å for HO···NH). The slight difference in the average distances result from enhanced degrees of freedom and the relaxation of the complex in the solution phase. Collectively, these results point to the stability of the dimeric assembly in organic media.

The ability of the present receptors to remove NaOH from an aqueous source phase was studied initially via LLE using 1 mM NaOH in deionized water and receptors *cis*-2 or *trans*-2 as 1 mM solutions in toluene. The measured pH decreased from 11.02 to 10.56 and 11.01 to 10.44 within 1 hour (Fig. 4a). Using equation 2,⁴⁸ these changes correspond to NaOH removal efficiencies of ca. 65% and 70% in the case of *cis*-2 and *trans*-2, respectively.

$$\text{E\%} = \Delta[\text{NaOH}]/[\text{NaOH}]_0 \times 100\% \quad (2)$$

Concurrently, a decrease in the conductivity of the alkaline phase, from 238 μS/cm to 90 μS/cm and from 238 μS/cm to

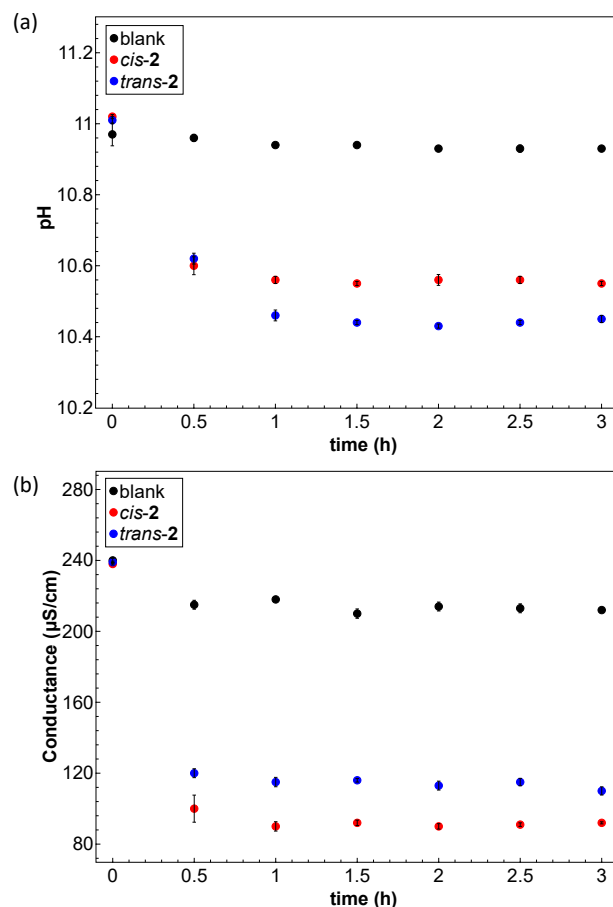


Figure 4. Time-dependent [NaOH] measurements of the aqueous layer of LLE by (a) pH; (b) conductance. Samples were prepared as follows: [NaOH] = 1 mM in D.I., [*cis*-2] and [*trans*-2] = 1 mM in toluene. A 0.5 mL of the aqueous solution was stirred with 0.5 mL of each organic test solution at 700 rpm under N₂ in poly(propylene) (PP) vials. The aqueous solution was then transferred for conductivity measurements, followed by treatment with 20 μL of concentrated KCl (200 mM in D.I.) so as to maintain an ionic strength 0.01 M for pH measurements. Error bars represent std. dev., n = 3.

110 μS/cm, was observed for *cis*-2 and *trans*-2, respectively (Fig. 4b). On the basis of a [NaOH]-conductivity calibration curve (*cf.* Fig. S10), these reductions in salinity correspond to NaOH removal efficiencies of 65% and 59% by *cis*-2 and *trans*-2, respectively. Some precipitation upon contacting *cis/trans*-2 with NaOH was seen under these experimental conditions. Note that 1 mM aqueous NaOH provides a basic medium (pH = 10.90) that is not high enough to promote significant pyrrole NH deprotonation based on the reported pK_a value of 16.5 for pyrrole in water.⁴⁹

In order to gain insight into the nature of the precipitate, the LLE was scaled up and the resultant suspensions were isolated for analysis. Although precautions were taken to prevent CO₂ adsorption into the NaOH solutions (as noted in the experimental section), NaHCO₃/Na₂CO₃ may still be present and interfere with NaOH binding. Therefore, both precipitates of *cis*-2•NaOH and *trans*-2•NaOH were analyzed by attenuated total reflection infrared spectroscopy (ATR-IR). The resulting spectra were characterized by an absence of

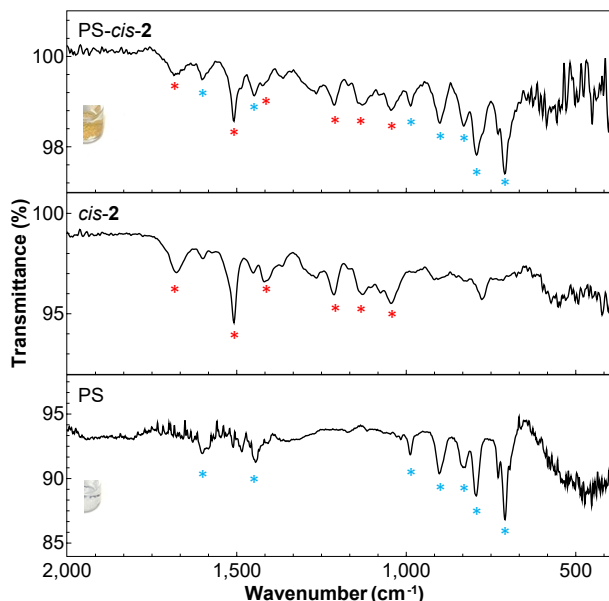


Figure 5. ATR-IR spectra of PS-*cis*-2, *cis*-2, and PS. Inset: pictures of beads in deionized water (D.I.).

characteristic vibrational features ascribable to NaHCO_3 or Na_2CO_3 (e.g., C=O stretching $\nu(\text{C=O})$ (1615 cm^{-1}) and C-O stretching $\nu(\text{C-O})$ ($993, 1923\text{ cm}^{-1}$) modes for NaHCO_3 , and C-O stretching $\nu(\text{C-O})$ (1431 cm^{-1}) and C-O bending $\delta(\text{C-O})$ (878 cm^{-1}) features for Na_2CO_3 , respectively). These observations along with the presence of vibrational features resembling those of NaOH was taken as evidence for the presumed NaOH capture in the case of both *cis*-2•NaOH and *trans*-2•NaOH (cf. Fig. S11).

Attempts to alleviate residual precipitation by further dilution proved unsuccessful. Therefore, an alternative approach was sought. Solvent impregnated resins (SIRs), wherein an extractant is immobilized onto a polymeric support via hydrophobic interactions, can provide an effective means of avoiding third phase formation, and have seen applications in both sensing and separations.^{50–54} We thus physically immobilized receptors *cis*-2 and *trans*-2 onto macroporous poly(styrene) resin Amberlite™ XAD-4 (denoted as PS hereafter) via solvent impregnation.⁵² When 15 mg of PS was exposed to 3 mg of receptors in 5 vol % (v/v) of acetonitrile ACN/ H_2O at 60°C for 1 h, the resins underwent a color change from opaque to amber (Figure 5 inset). This finding was taken as evidence that the receptors were being absorbed into the PS. Attempts to increase the receptor/PS loading ratio to 1:5 receptor:PS level produced a cloudy solution reflecting the presumed presence of non-adsorbed receptors in the medium.

The resins obtained as above were first characterized by ATR-IR spectroscopy. Characteristic IR absorptions of *cis*-2 and PS were observed for PS-*cis*-2 (Figure 5). Particularly notable were bands ascribable to C=O stretching $\nu(\text{C=O})$ (1681 cm^{-1}), C=C bending $\delta(\text{C=C})$ (1510 cm^{-1}), C-H bending $\delta(\text{C-H})$ (1420 cm^{-1}), C-O stretching $\nu(\text{C-O})$ (1213 cm^{-1}), and C-N stretching $\nu(\text{C-N})$ ($1130, 1040\text{ cm}^{-1}$) marked by red asterisks for *cis*-2, as well as C=C stretching $\nu(\text{C=C})$ ($1602, 1444\text{ cm}^{-1}$), and C-H bending $\delta(\text{C-H})$ ($991, 904, 830, 800, 710\text{ cm}^{-1}$) marked by blue asterisks for PS in Figure 5. Similar results were seen for PS-*trans*-2 (cf. Fig. S12).

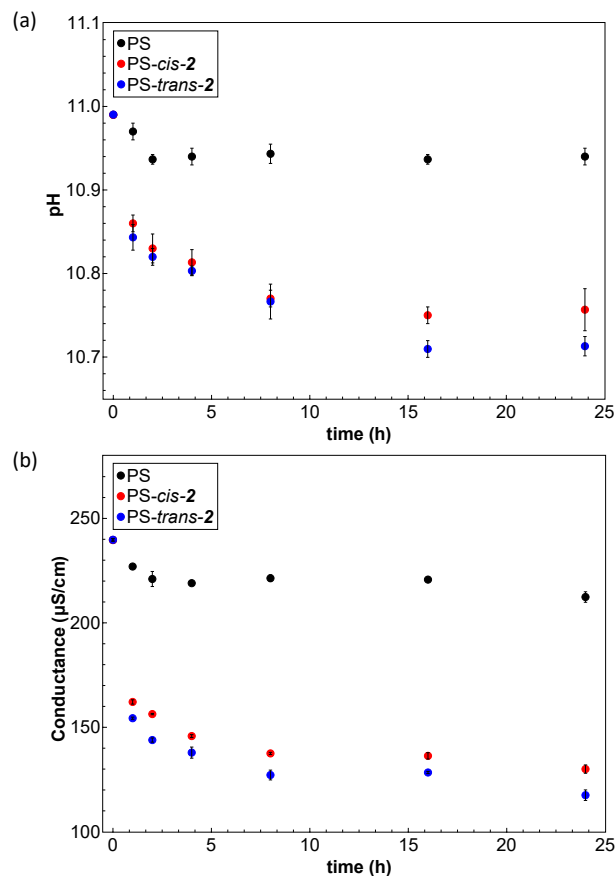


Figure 6. Time-dependent [NaOH] change of the aqueous phase extracted with SIRs and measured by (a) pH; (b) conductance. Samples were prepared as follows: $[\text{NaOH}] = 1\text{ mM}$ in D.I. A 1.5 mL of the NaOH solution was stirred with 15 mg of XAD-4 (PS), 17 mg of PS-*cis*-2, and 17 mg of PS-*trans*-2 at 700 rpm under N_2 in PP vials. The aqueous solution was then transferred for conductivity measurements, followed by treatment with 75 μL of concentrated KCl (200 mM in D.I.) so as to maintain an ionic strength 0.01 M for pH measurements. Error bars represent std. dev., $n = 3$.

Next, quantification of receptor loading was assessed by UV-vis studies of desorption of receptors *cis*-2 and *trans*-2. In THF, the receptors were desorbed from PS-*cis*-2 and PS-*trans*-2 effectively as inferred from the discoloration of the PS resin. Both THF solution obtained in this way showed UV-vis spectra characteristic of *cis*-2 and *trans*-2 in 4% by volume ACN/ H_2O (cf. Fig. S14). The mass of the desorbed receptors could be determined from the absorbance (0.227, 0.259 for PS-*cis*-2 and PS-*trans*-2), solution volumes (5 mL), dilution factors ($\times 25$), and extinction coefficients ($\epsilon = 10477$ and extinction coefficients ($12147\text{ M}^{-1}\cdot\text{cm}^{-1}$ at 285 nm for *cis*-2 and *trans*-2, respectively). On average, 2.5 mg of *cis*-2 and 2.5 mg of *trans*-2 were desorbed from 15 mg of PS-*cis*-2 and PS-*trans*-2, corresponding to 16.8 wt % and 16.5 wt % loadings.

Extraction of NaOH by the SIRs PS-*cis*-2 and PS-*trans*-2 was then studied by both pH and conductivity measurements. While only a minute decrease from 10.99 to 10.94 in pH was seen for the control PS, PS-*cis*-2 and PS-*trans*-2 produced changes from 10.99 to 10.75 and 10.99 to 10.71 over a 24 h measurement period (Fig. 6a). Such a decrease corresponds to NaOH removal efficiencies, calculated according to equation

Table 2. Capacities and rate constants of NaOH adsorption by PS, PS-*cis*-2 and PS-*trans*-2 derived from pH and conductance measurements.*

	pH		Conductance	
	Q_{eq} (meq/g)	k (g·meq ⁻¹ ·h ⁻¹)	Q_{eq} (meq/g)	k (g·meq ⁻¹ ·h ⁻¹)
PS	0.009	155.1	0.010	140.3
PS- <i>cis</i> -2	0.031	42.8	0.045	48.3
PS- <i>trans</i> -2	0.034	35.4	0.049	47.9

*Data were fitted to the pseudo-second order equation: $\frac{t}{Q_{eq}} = \frac{t}{Q_{eq}^2} + \frac{1}{kQ_{eq}^2}$

2, of 42% and 47% for PS-*cis*-2 and PS-*trans*-2, respectively. Likewise, the conductivity of the NaOH solutions decreased from 240 μ S/cm to 130 μ S/cm for PS-*cis*-2 and 118 μ S/cm for PS-*trans*-2, corresponding to NaOH removal efficiencies of 44% and 49%, respectively. As anticipated, no precipitation was seen. Their lower NaOH removal efficiencies in comparison to free-flowing receptors *cis*-2 and *trans*-2 may be attributable to the reduced mobility. The sandwiched structures permitted by the crown ethers contribute to a higher binding affinity for Na⁺. Once immobilized, the receptors are constrained such that the probability of adopting sandwiched structures is reduced. As a result, lower efficiencies are seen.

The equilibrium capacities Q_{eq} (equiv of NaOH adsorbed by the resin at equilibrium) and rate constants of NaOH adsorption by the present SIRs could be determined by fitting the kinetic data to a pseudo-second order adsorption model (cf. Fig. S15).⁵⁵ The results are presented in Table 2. Overall, a good concordance was found between the pH and conductance studies, although in general higher values were obtained from the conductance measurements. This difference might reflect intrinsic alkaline errors associated with measuring low [H⁺] solutions.⁵⁶ The longer equilibrium time required for the pH electrode compared to conductivity electrodes (>15 s vs <5 s) could also introduce errors caused by CO₂ absorption into the test solutions. Critically, both PS-*cis*-2 and PS-*trans*-2 demonstrated substantial increase in Q_{eq} , whereas unmodified PS did not. Taking the conductance data by way of example, this value increased by 0.035 meq/g and 0.039 meq/g with respect to pristine PS. Both PS-*cis*-2 and PS-*trans*-2 also displayed comparable adsorption kinetics, with $k = 48.3$ g·meq⁻¹·h⁻¹ and 47.9 g·meq⁻¹·h⁻¹ being seen for PS-*cis*-2 and PS-*trans*-2, respectively. Of note is that no appreciable change in the ATR-IR spectra of PS-*cis*-2 and PS-*trans*-2 is seen upon contact with NaOH leading to the inference that these species are stable under the conditions of our experiments (cf. Fig. S13).

An inherent concern for SIRs is leaching of the adsorbed extractant over prolonged use.⁵² We thus subjected the NaOH solution extracted with SIRs to UV-vis spectral analyses over a 24 h monitoring period (cf. Fig. S16). Taking into account the solution volume (1.5 mL) and dilution factor ($\times 3$), the individual absorbance values were converted to mass. The highest mass loss, 0.027 mg and 0.017 mg *cis*-2 and *trans*-2, respectively, were found after 24 h. These values correspond to only 1 wt % and 0.7 wt %, leaching, with respect to the initial loading levels (2.5 mg of receptors onto 15 mg of PS) (cf. Fig. S17).

Conclusion

In this work we demonstrated that pH can be lowered by active removal of NaOH by ion-pair receptors *cis*-2 and *trans*-2, calix[4]pyrroles appended with two benzo-12-crown-4 ethers. A dimerized structure of [*cis*-1•NaOH]₂ was found in the solid-state as evidenced by single crystal structure, whereas *cis*-2 and *trans*-2 dimerize and trimerize with NaOH in toluene as inferred from DOSY studies. DFT and AIMD simulations support the stability of the dimeric supramolecular assemblies relative to the corresponding 1:1 complexes. Although the propensity of these receptors to precipitate NaOH likely precludes their use as extractants under LLE conditions, immobilization onto macroporous poly(styrene) supports enables direct NaOH removal from a high pH aqueous source phase. We thus suggest that *cis*-2 and *trans*-2 or analogous systems capable of capturing NaOH effectively may have a role to play in lowering the pH of basic aqueous media without increasing salinity or the original solution volumes.

ASSOCIATED CONTENT

Supporting Information.

¹H and ¹³C NMR, ESI-MS analyses of compounds; DOSY spectra of *cis*/*trans*-2 and *cis*/*trans*-2•NaOH; ATR-IR spectra of PS-*trans*-2, [NaOH]-conductivity calibration curve; UV-vis spectra of *cis*-2, *trans*-2, desorbed receptors from PS-*cis*-2 and PS-*trans*-2; kinetic fitting; X-ray experimental; computational methods and analyses.

This material is available free of charge via the Internet at <http://pubs.acs.org>.

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ABBREVIATIONS

EDC, 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide; HOBt, hydroxybenzotriazole; DIPEA, *N,N*-diisopropylethylamine;

LLE, liquid-liquid extraction; DOSY, diffusion-ordered spectroscopy; SIRs, solvent impregnated resins; PP, poly(propylene); PS, macroporous poly(styrene) resin Amberlite™ XAD-4; E%, extraction efficiency; Q_{eq} , equilibrium capacity; IEFPCM, integral equation formalism variant of the polarizable continuum model.

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